```c
#include<stdio.h>
#include<stdlib.h>
int constantspace(int n)
{
    int a=20;
    int b=70;
    int sum=a+b+n;
    return sum;
}
int main()
{
    //O(1)
    int n=1000;
    printf("O(1) space complexity example \n");
    int result=constantspace(n);
    printf("result=%d\n",result);
    printf("space used=%lu bytes\n",sizeof(int)*6); //6 variables used

    //O(n)
    printf("O(n) space complexity example \n");
    int m;
    printf("enter no of elements in the array \n");
    scanf("%d",&m);
    int *arr=(int *)malloc(n *sizeof(int));
    for(int i=0;i<=m;i++)
    {
        arr[i]=i;
    }
    printf("space used=%lu bytes \n",m*sizeof(int));
    free(arr);

    //O(n^2)
    printf("O(n^2) space complexity example \n");
    int p;
    printf("enter no of rows in the matrix \n");
    scanf("%d",&p);
    int**matrix=(int **)malloc(p*sizeof(int*));
    for(int i=0;i<=p;i++)
    {
        matrix[i]=(int *)malloc(p* sizeof(int));
    }
    for(int i=0;i<=p;i++)
    {
        for(int j=0;j<=p;j++)
```

```
        {
            matrix[i][j]=i+j;
        }
    }
    printf("space used =%lu bytes \n",p*p*sizeof(int));
    for(int i=0;i<=p;i++)
    {
        free(matrix[i]);
    }
    free(matrix);
    return 0;

}
```

Output

```
O(1) space complexity example
result=1090
space used=24 bytes
O(n) space complexity example
enter no of elements in the array
5
space used=20 bytes
O(n^2) space complexity example
enter no of rows in the matrix
5
space used =100 bytes
```